Internet of Things

By

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Outline

- IoT Foundations: Low-Power Wireless Networking
- When to use
- Low power Wide Area Network
- LTE-based IoT wireless networks
- LoRaWAN for IoT wireless networks
- Integrate wireless networking options
- 5G technology

Key features of 6LoWPAN

- Worldwide and regional unlicensed ISM bands
- Various network topology options: star, tree, and mesh
- Support for low-power, resource-constrained end devices
- IEEE 802.15.4-based security schemes
- IPv6 support for IEEE 802.15.4-based wireless networks

IPv6 Support

- Suitable for dense LoWPANs
- Does not require proxies for connecting to other IP networks
- Reusing protocols and tools with IP networks

6LoWPAN network

6LoWPAN Host (6LH)

End devices that do not route packets

6LoWPAN Border Router (6LBR)

Manage the network and handle the protocol translation

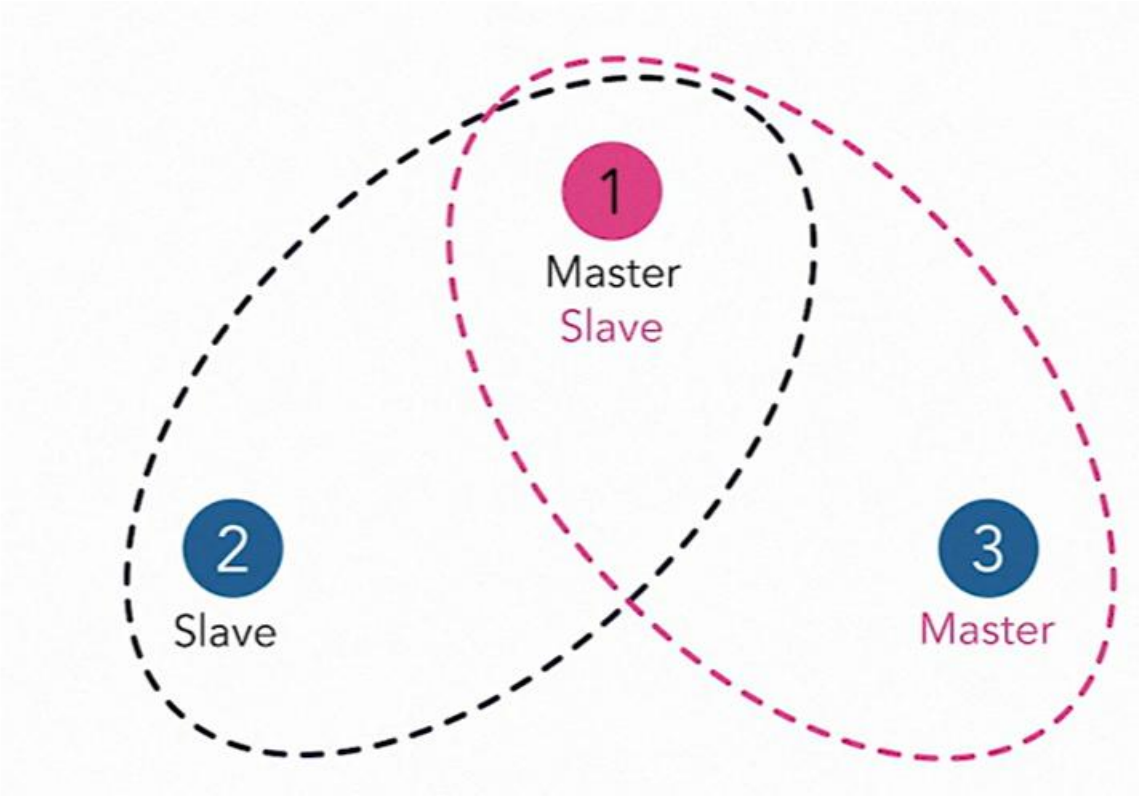
Bluetooth low energy for IoT Wireless networks

- Worldwide unlicensed ISM band at 2.4 GHz
- Support multiple physical-layer and topology options
- Multiple security options
- IP support is available

Applications

- Smart building
- Smart home
- Healthcare
- Computer peripherals and accessories
- Data transfer

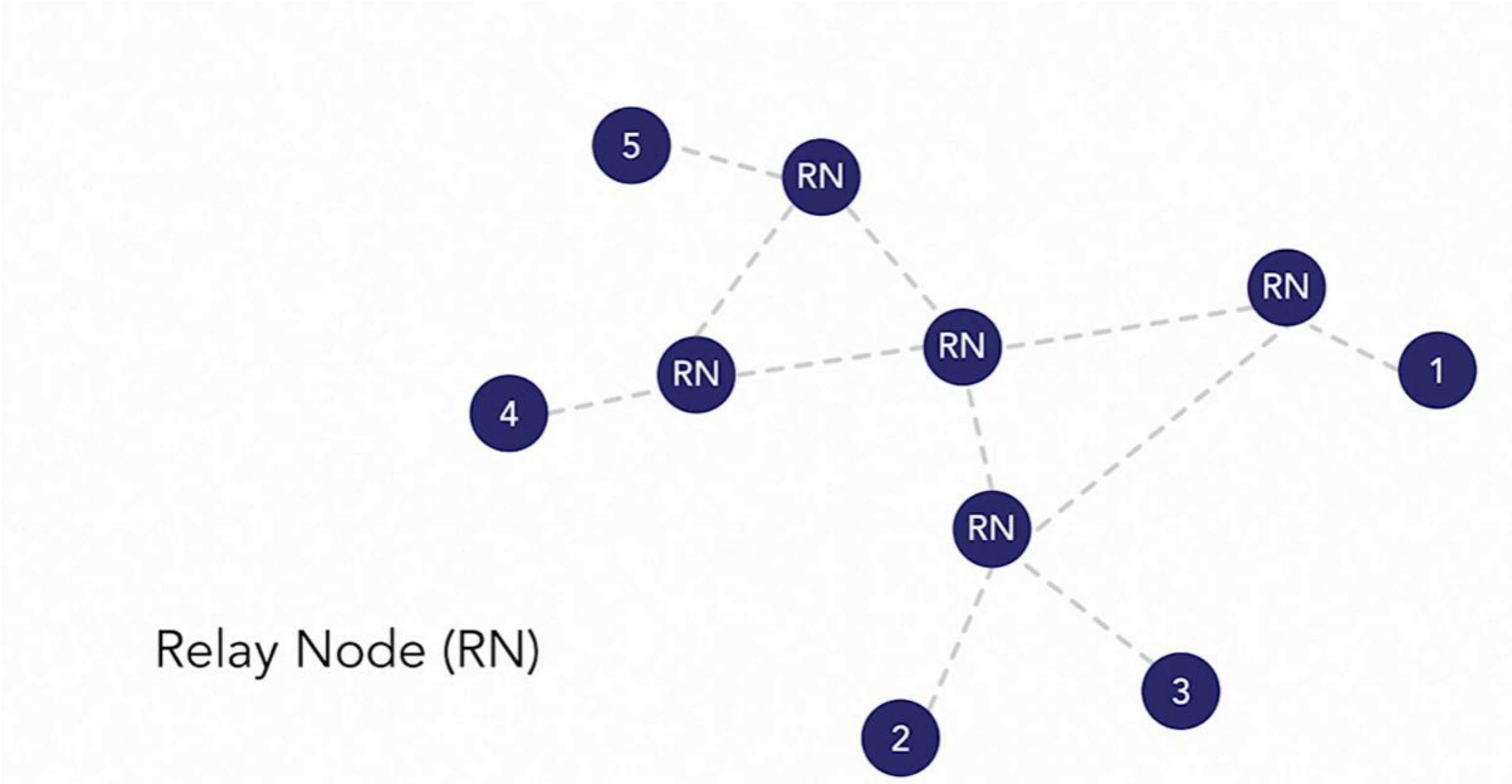
Network Architecture



Scattered net

A "scatternet" is an ad hoc network formed by connecting two or more piconets (independent Bluetooth networks)

Blue tooth network



A relay node, in networking, is an intermediary device that receives and forwards data packets between different parts of a network, acting as a crucial point for data transmission when direct communication is not feasible

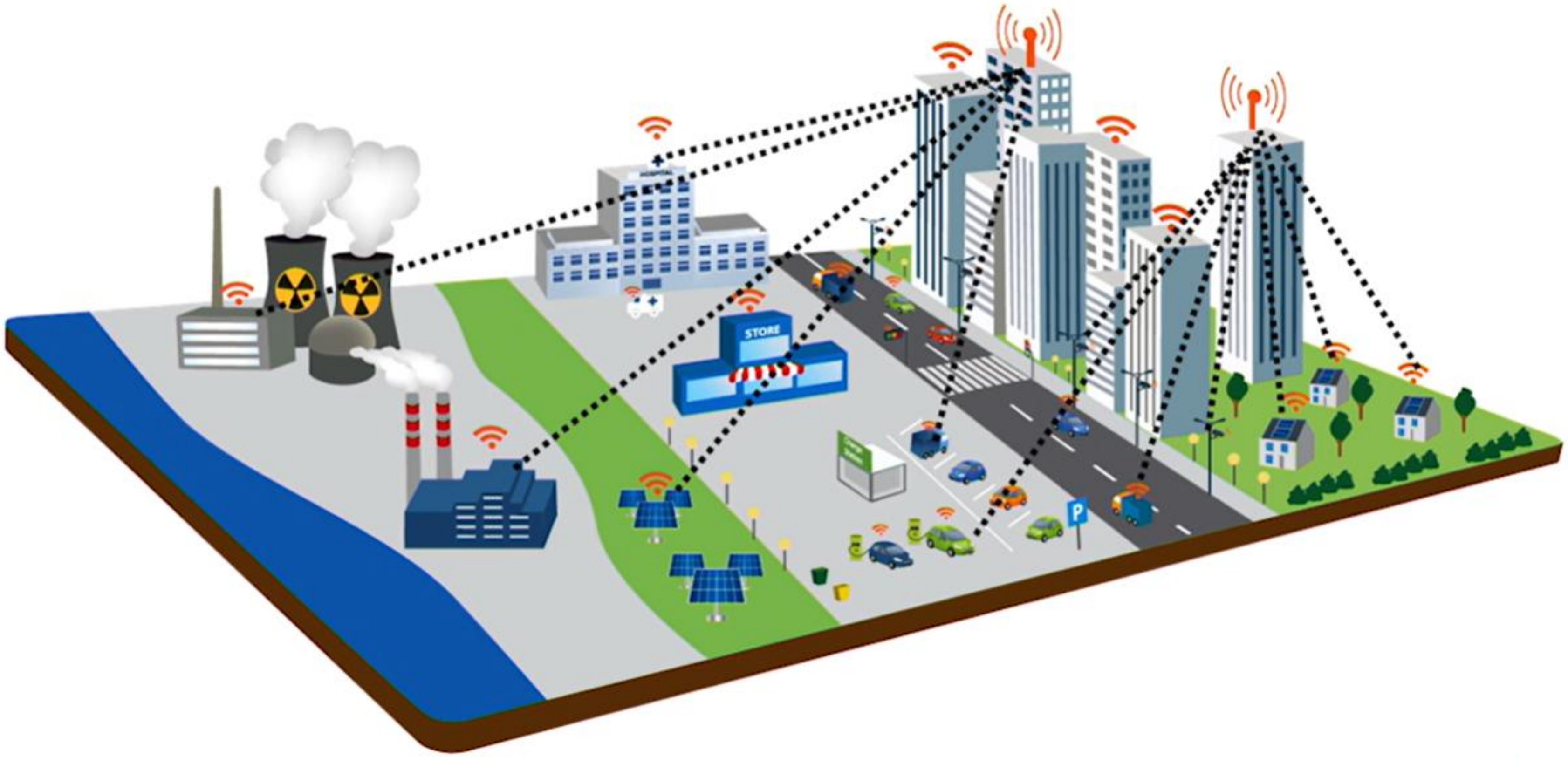
Platforms

- Bluetooth stacks on Windows, macOS, iOS, and Android
- BlueZ on Linux distributions
- Zephyr for embedded systems
- Bluetooth SDKs on hardware platforms

LPWAN features

- Provide a long-range, wide-area coverage
- Provide a long battery life on devices
- Support low-cost applications
- Connect massive amounts of end devices
- Support various traffic patterns

Example use cases



Example Use cases

Smart city

- Smart street lighting
- Smart parking

Smart metering

- Electricity metering

Manufacturing automation

- Machine condition monitoring
- Process automation

Example Use cases

Cellular LPWAN Technologies

- EC-GSM-IoT
- LTE-M
- NB-IoT

Noncellular LPWAN Technologies

- LoRa
- SIGFOX

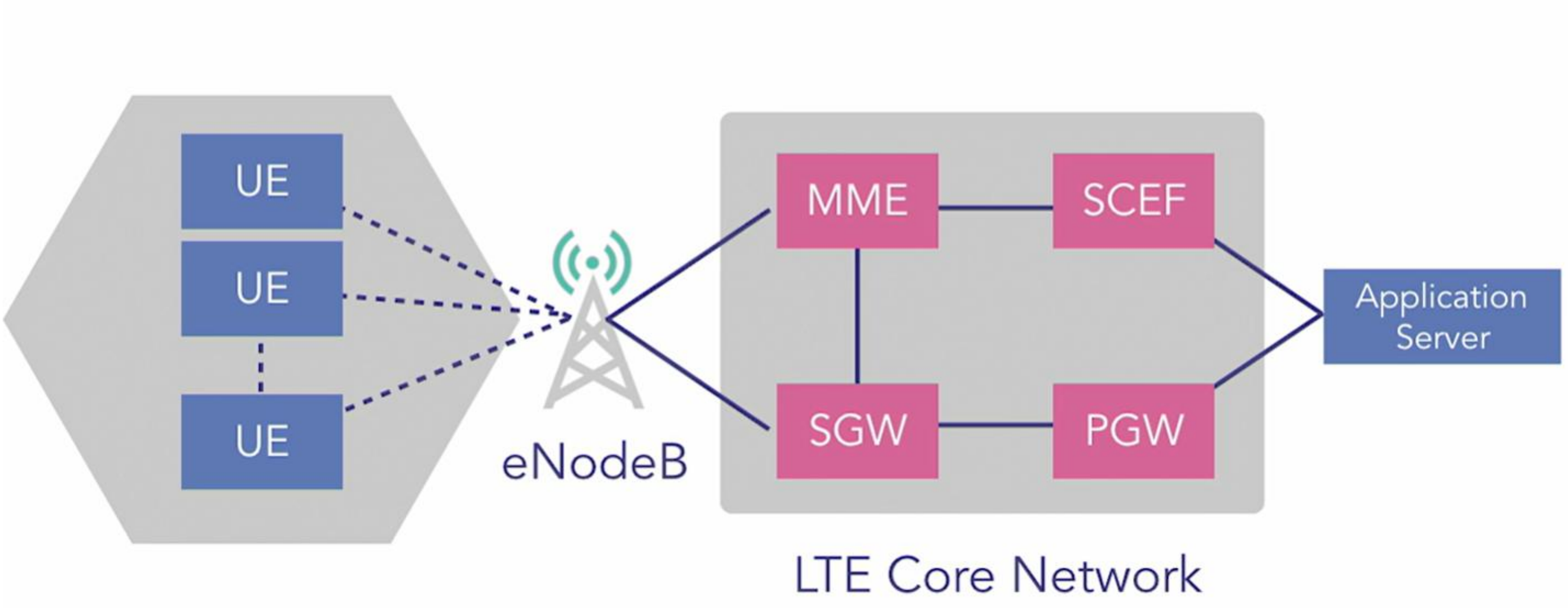
Comparison of LPWAN

Parameters	LoRa	EC-GSM-IoT	LTE-M	NB-IoT
Radio spectrum	Unlicensed sub-1 GHz	Licensed GSM bands	Licensed LTE bands in-band	Licensed LTE in-band, guard band & standalone
Downlink data rate	<0.05 Mbps	<0.14 Mbps	<1 Mbps	<0.17 Mbps
Uplink data rate	<0.05 Mbps	<0.14 Mbps	<1 Mbps	<0.25 Mbps
Range	<15 km	<35 km	<100 km	<35 km

LTE based LPWAN

- Support massive IoT devices
- Provide wide coverage
- Use low-power devices
- Enable low-cost applications
- Reuse existing LTE infrastructure

Network Architecture



User equipment

serving gateway (SGW)-data packet transfer

PGW packet data gateway—providing connectivity

MME manages mobility and session control for LTE networks, while SCEF exposes network capabilities for non-IP data delivery and other services.

User Equipment category

- LTE Cat 1
- LTE Cat 0
- LTE Cat-M1 (LTE-M)
- LTE Cat-NB1 (NB-IoT)

Category 1 can transmit data within several mbps , broadband system to mobile network, cat 0- 1mbps data rate uplink and downlink, but much lower power consumption, CATM-1.4mbps data rate. complexity reduction and power saving, NB has further complexity and power reduction.

Technical comparison

Parameters	LTE Cat 0	LTE Cat 1	LTE Cat-M1	LTE Cat-NB1
Spectrum	Licensed LTE in-band	Licensed LTE in-band	Licensed LTE in-band	Licensed LTE in-band, guard band
Downlink rate	<1 Mbps	<10 Mbps	<1 Mbps	<0.17 Mbps
Uplink rate	<1 Mbps	<5 Mbps	<1 Mbps	<0.25 Mbps
Transmit power	Low	Low	Low	Low

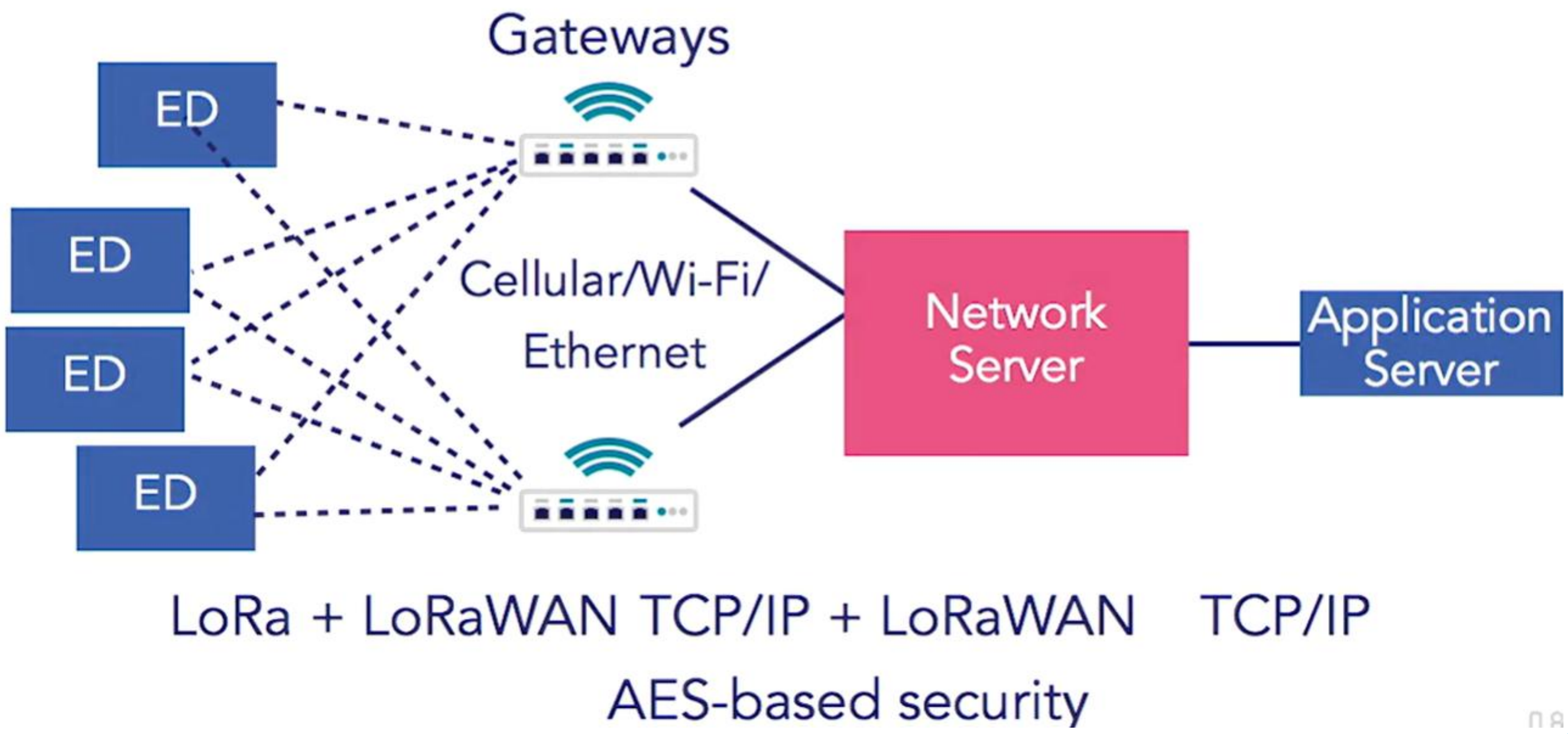
LoRa

- LoRa is the physical-layer technology for LPWANs.
- LoRaWAN is a network protocol specification for battery-powered IoT solutions standardized by the LoRa Alliance. It works with the LoRa technology to create a long-range communication capability.

LoRa based LPWAN

- Long-range, low-power, low-rate communication
- Support for regional unlicensed frequency bands
- Adaptable data rates
- Star-of-stars topology
- Support private, public, or hybrid networks

Network Architecture



Classes of End devices

Class A end devices

Simply transmit anytime and then remain inactive

Class B end devices

Transmit and listen in an assigned time slot

Class C end devices

Continuously listen and transmit

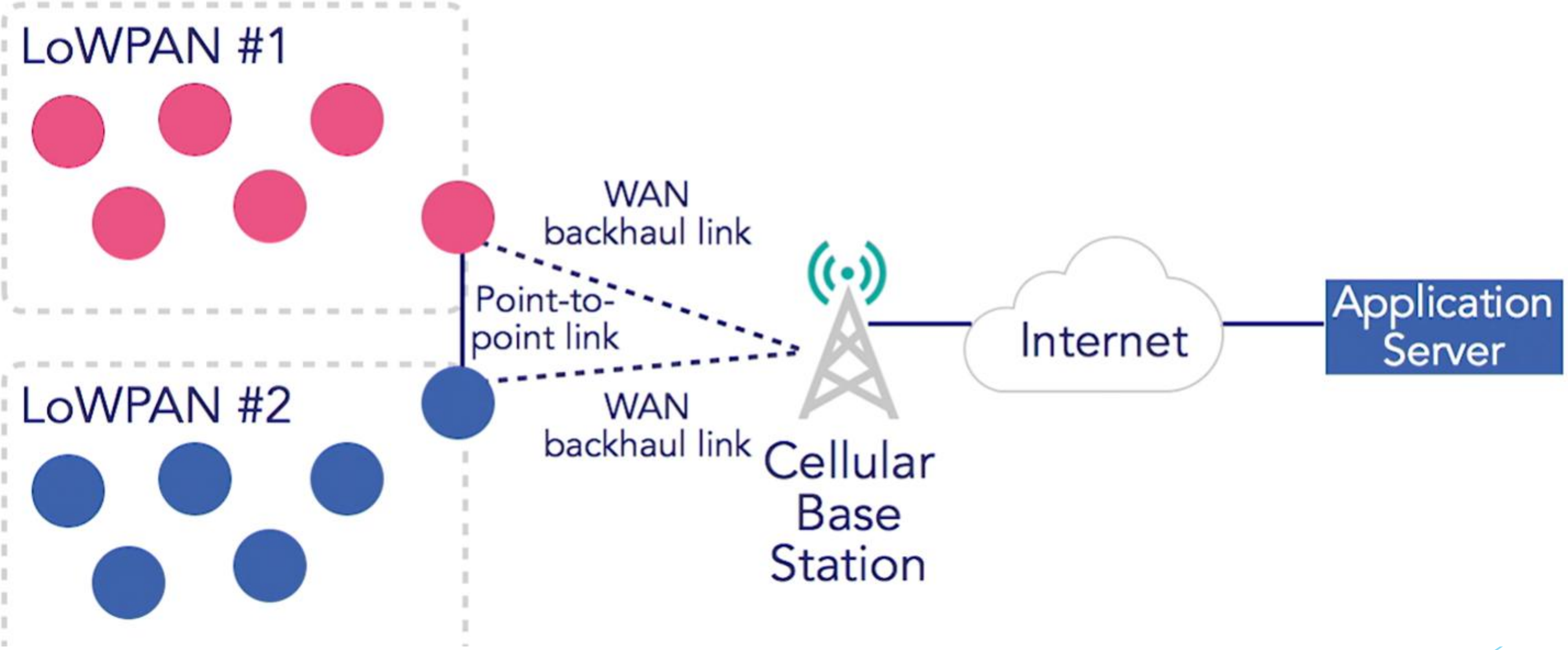
Example Target Application

- Precision agriculture
- Manufacturing automation
- Building automation

Necessity of System integration

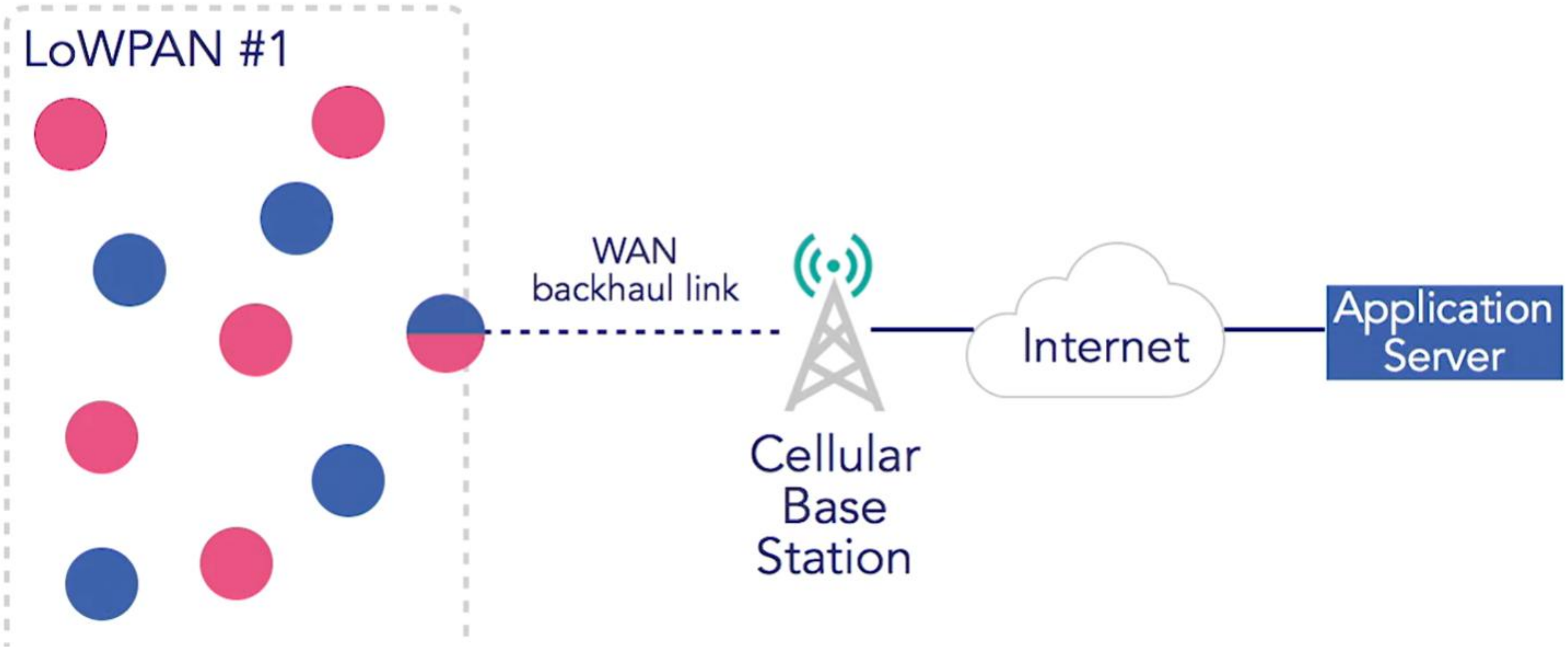
- Address heterogeneity of wireless networks
- Upgrade existing IoT systems
- Make IoT solutions scalable and interoperable

Scenario 1



LoWPAN #1
WAN
backhaul link
Point-to-
point link
LoWPAN #2
WAN
backhaul link
Cellular
Base
Station
Internet
Application
Server

Scenario 2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WAN
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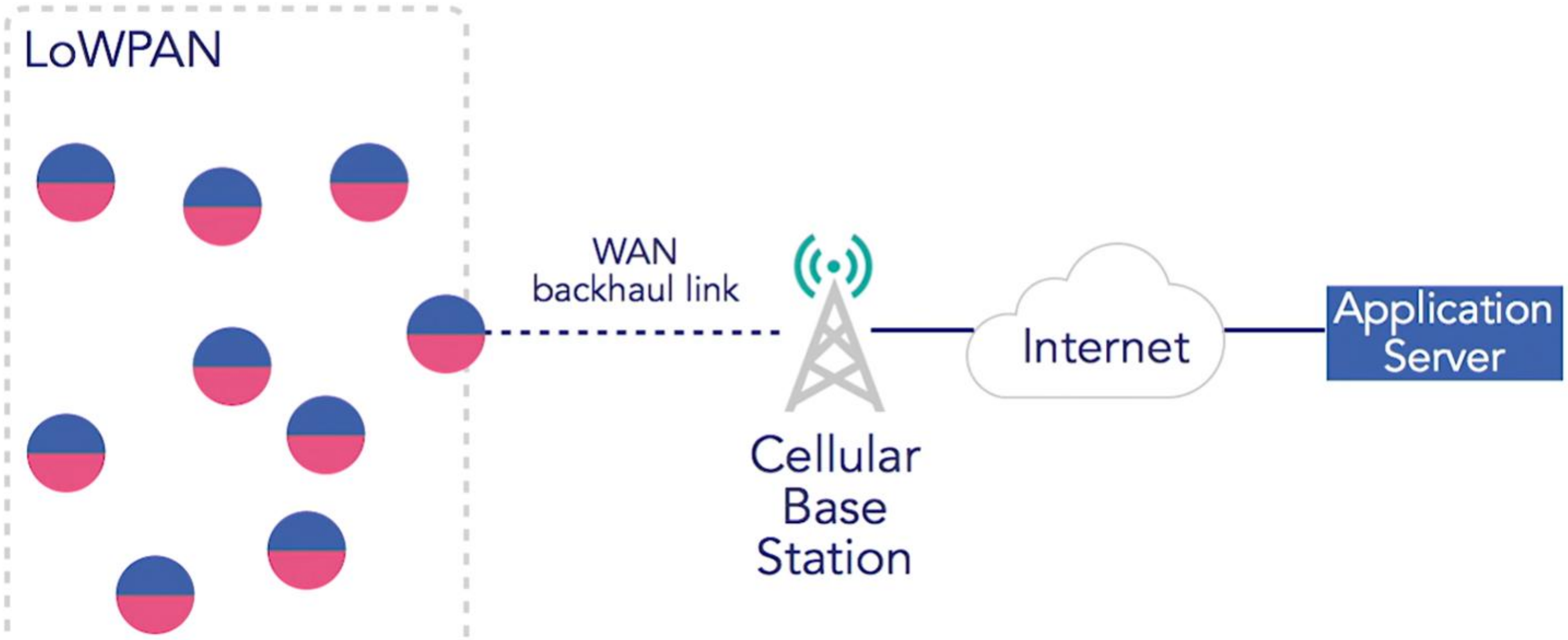
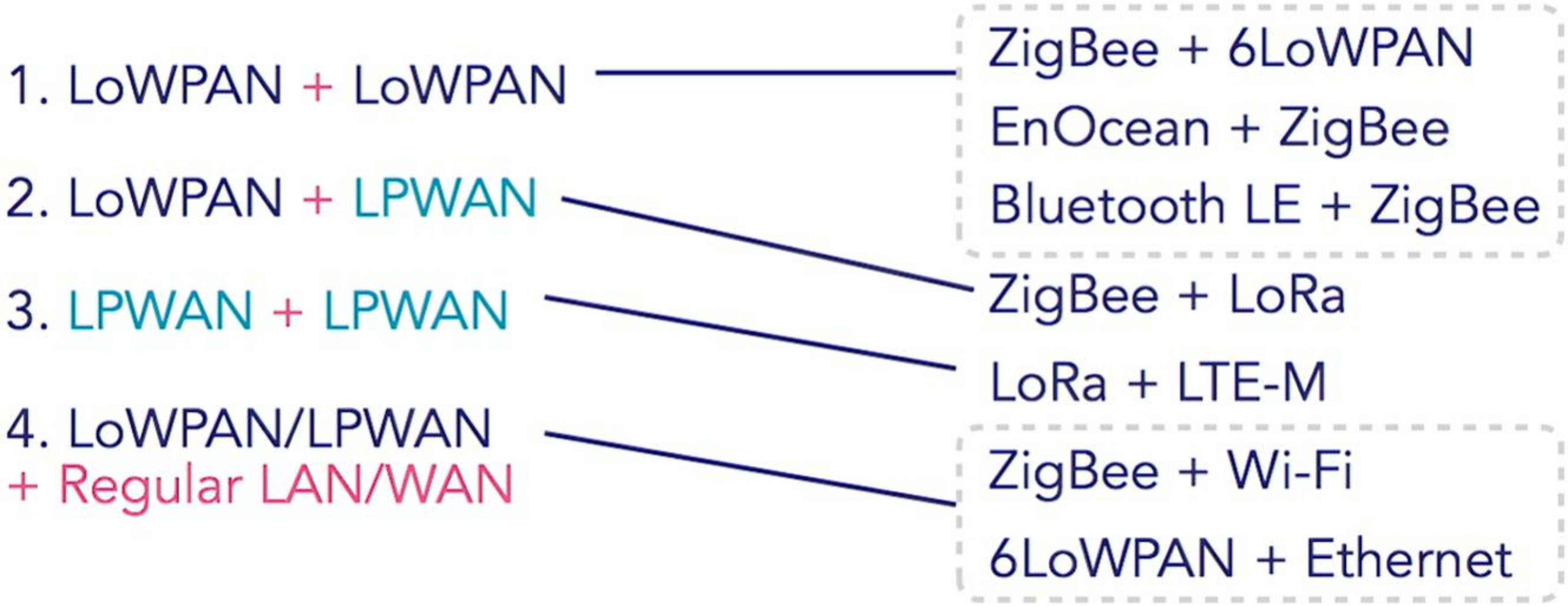
Scenario 2



LoWPAN
WAN
backhaul link
Cellular
Base
Station
Internet
Application
Server

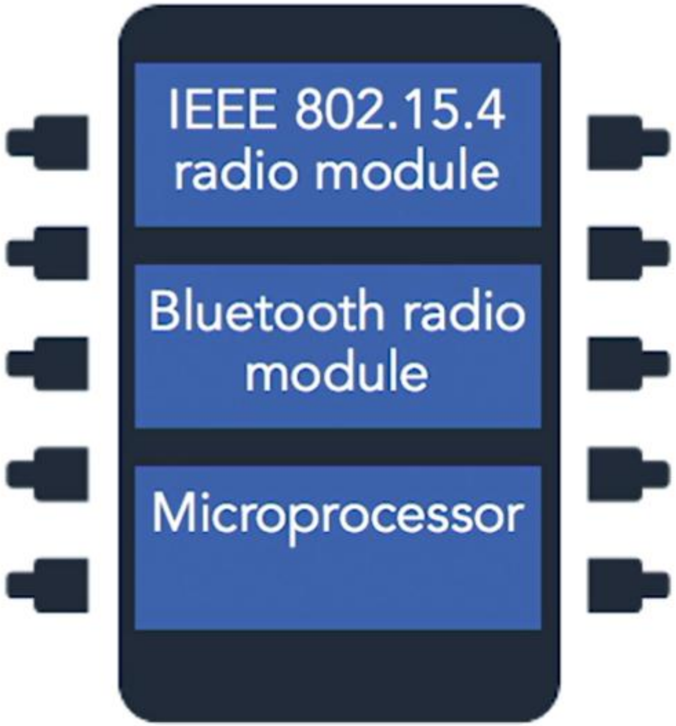
Typical integration options



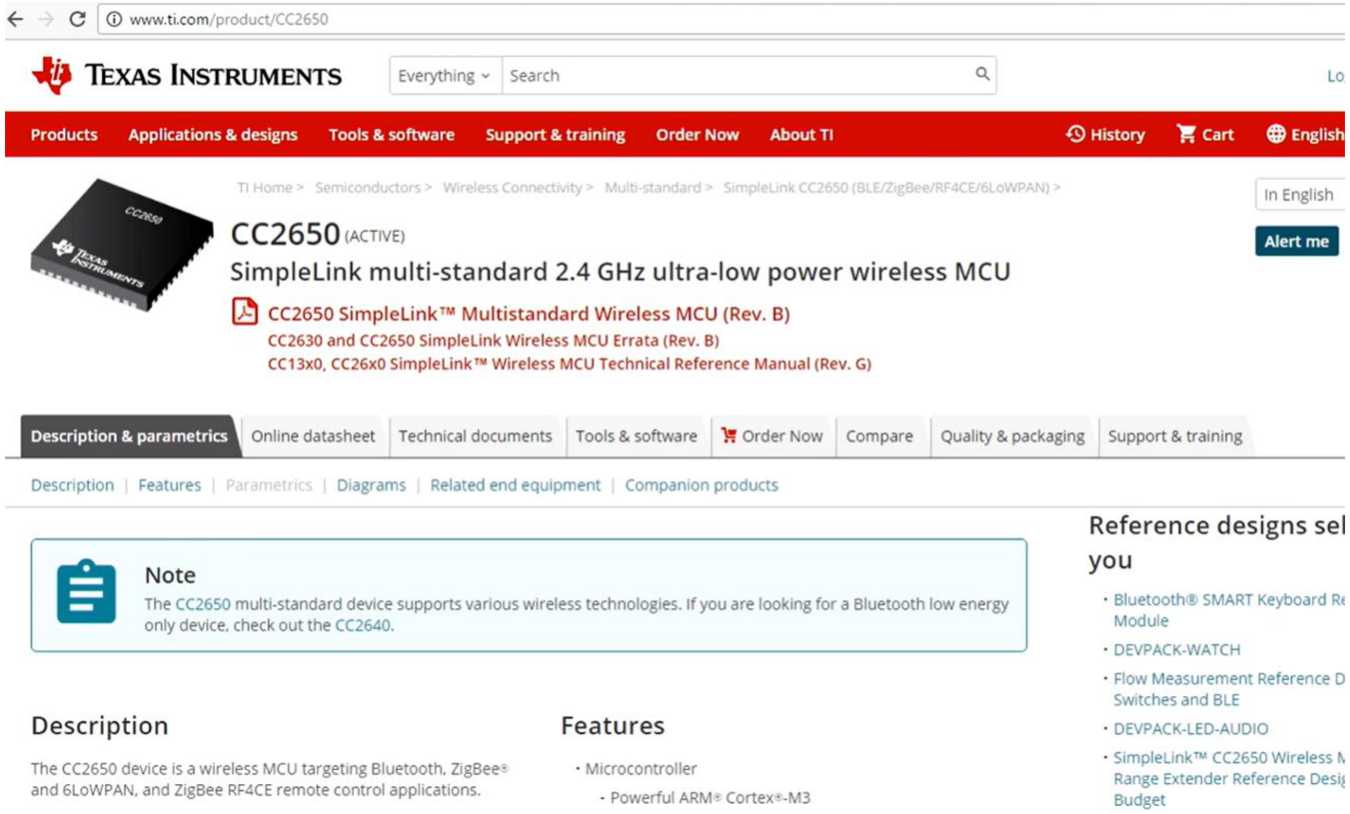
Chip level integration

Single microchip supporting

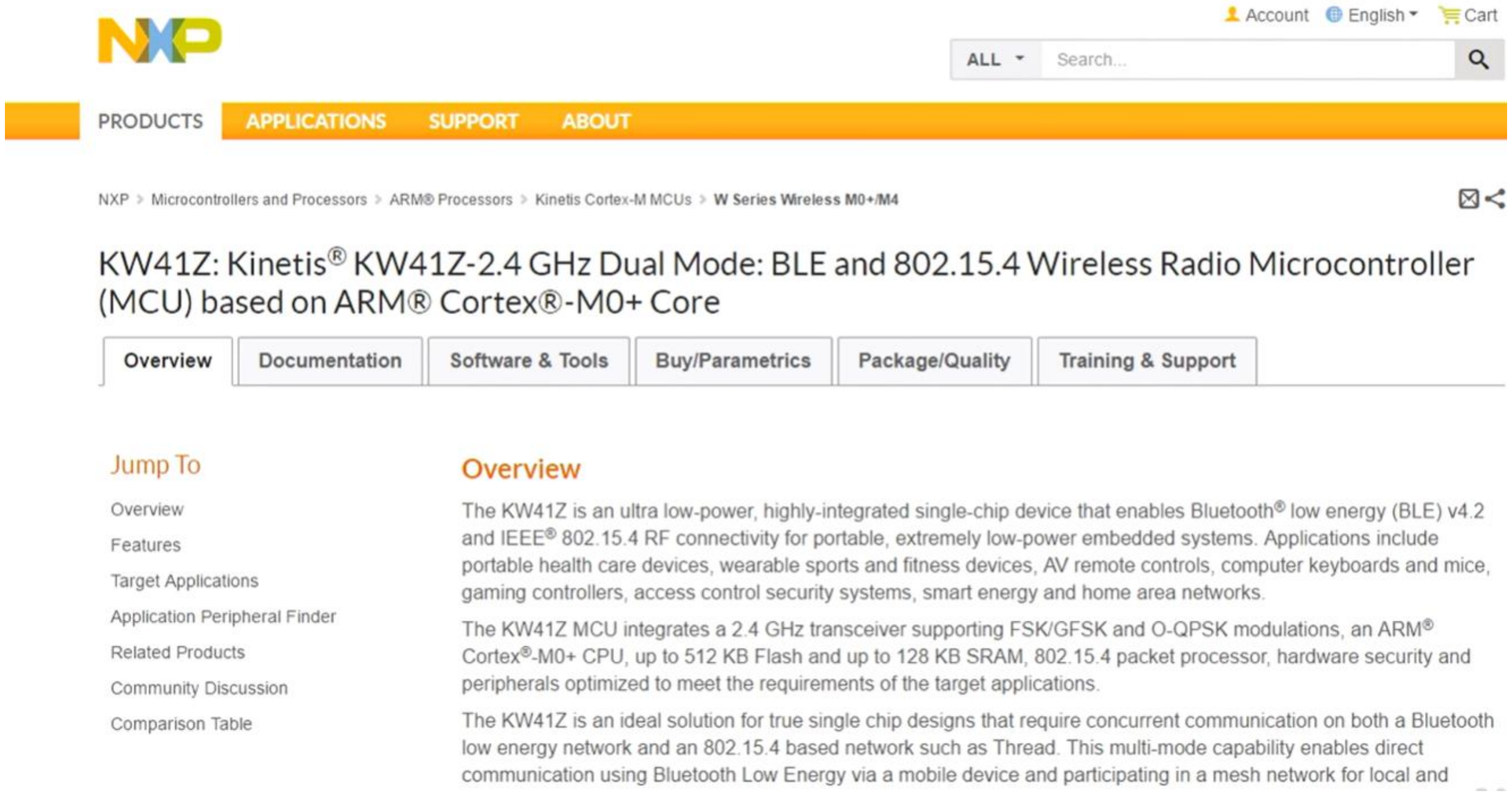
multi-standard, low-power wireless protocols



Single chip solution



Single chip solution



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PRODUCTS APPLICATIONS SUPPORT ABOUT

NXP > Microcontrollers and Processors > ARM® Processors > Kinetis Cortex-M MCUs > **W Series Wireless M0+/M4**

KW41Z: Kinetis® KW41Z-2.4 GHz Dual Mode: BLE and 802.15.4 Wireless Radio Microcontroller (MCU) based on ARM® Cortex®-M0+ Core

Overview | Documentation | Software & Tools | Buy/Parametrics | Package/Quality | Training & Support

Jump To

- Overview
- Features
- Target Applications
- Application Peripheral Finder
- Related Products
- Community Discussion
- Comparison Table

Overview

The KW41Z is an ultra low-power, highly-integrated single-chip device that enables Bluetooth® low energy (BLE) v4.2 and IEEE® 802.15.4 RF connectivity for portable, extremely low-power embedded systems. Applications include portable health care devices, wearable sports and fitness devices, AV remote controls, computer keyboards and mice, gaming controllers, access control security systems, smart energy and home area networks.

The KW41Z MCU integrates a 2.4 GHz transceiver supporting FSK/GFSK and O-QPSK modulations, an ARM® Cortex®-M0+ CPU, up to 512 KB Flash and up to 128 KB SRAM, 802.15.4 packet processor, hardware security and peripherals optimized to meet the requirements of the target applications.

The KW41Z is an ideal solution for true single chip designs that require concurrent communication on both a Bluetooth low energy network and an 802.15.4 based network such as Thread. This multi-mode capability enables direct communication using Bluetooth Low Energy via a mobile device and participating in a mesh network for local and

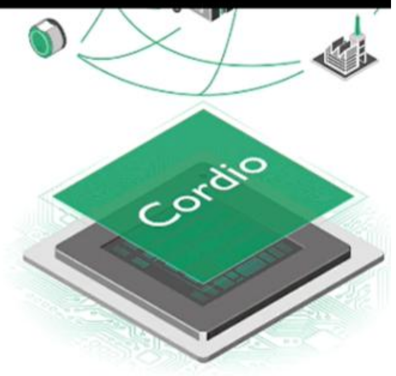
Single chip solution

ARM COMPANY MARKETS PRODUCTS DEVELOP NEWS

Cordio IP

Processors | Graphics and Multimedia | Physical IP | Development Tools | IoT Solutions | Security on ARM | System IP | Wireless IP

Request information for LPWAN solutions



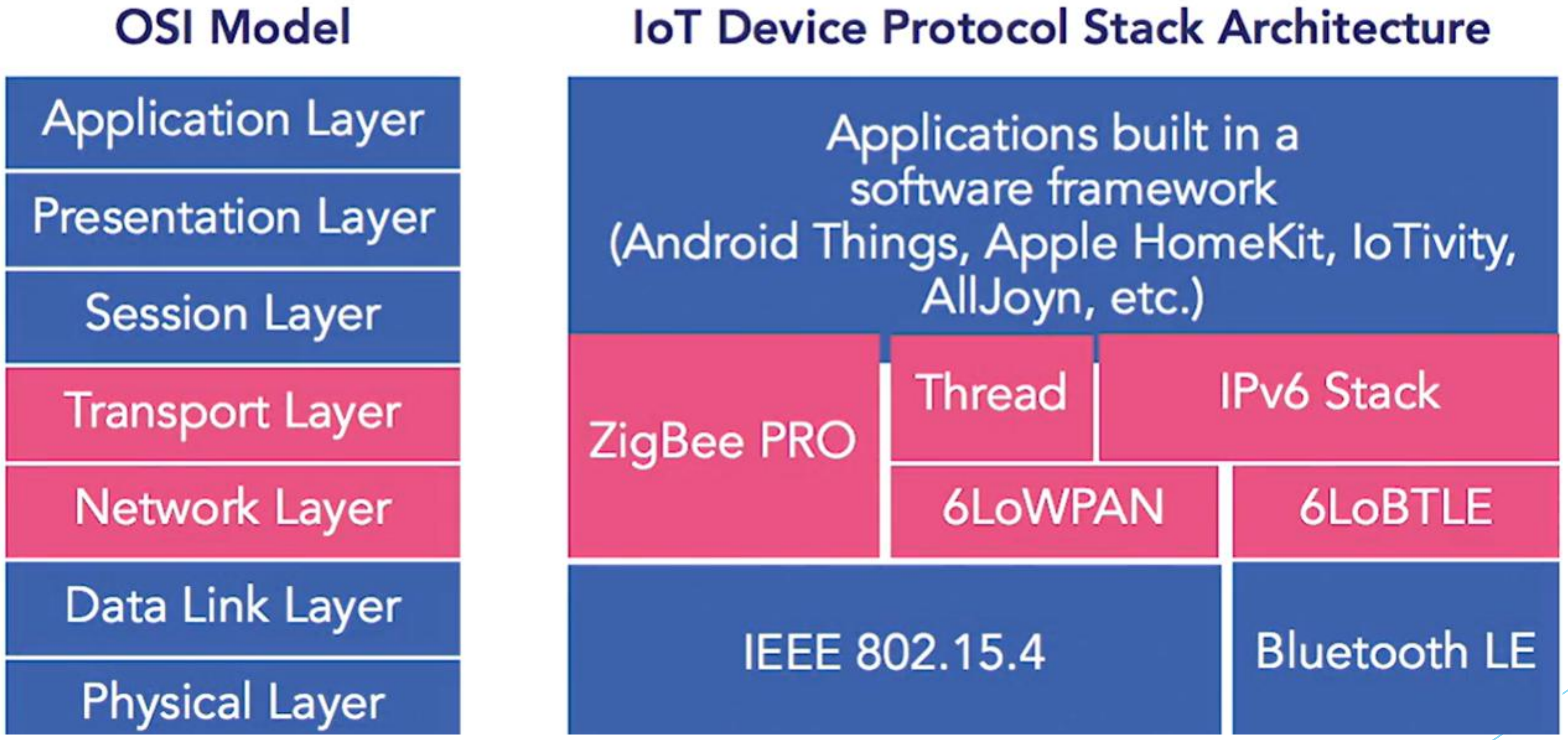
WPAN – Bluetooth 5 and 802.15.4

ARM Cordio IP provides on-chip radio connectivity for SoCs implementing WPAN wireless solutions for IoT. Cordio radio cores are fully integrated solutions with a hardware core and software stacks, and profiles. These low-voltage, low-power cores are designed for optimal efficiency with the ARM Cortex-M processor family.

The Cordio architecture brings extensive choice and flexibility to SoC designers. It provides foundry/process support, radio standards, and software stacks for easy integration to their SoC, as well as expanded support through third parties. With the ARM Cordio family of radio IP, semiconductor companies have access to cutting-edge, sub-volt radio solutions and can effectively keep up with evolving standards and processes in a cost-effective, risk-free manner. ARM Cordio portfolio consists of complete sub-volt radio IP solutions (hardware sub-system), radio standards delivered as soft IP and software (Link/MAC, stack and profiles).

Request information for WPAN solutions

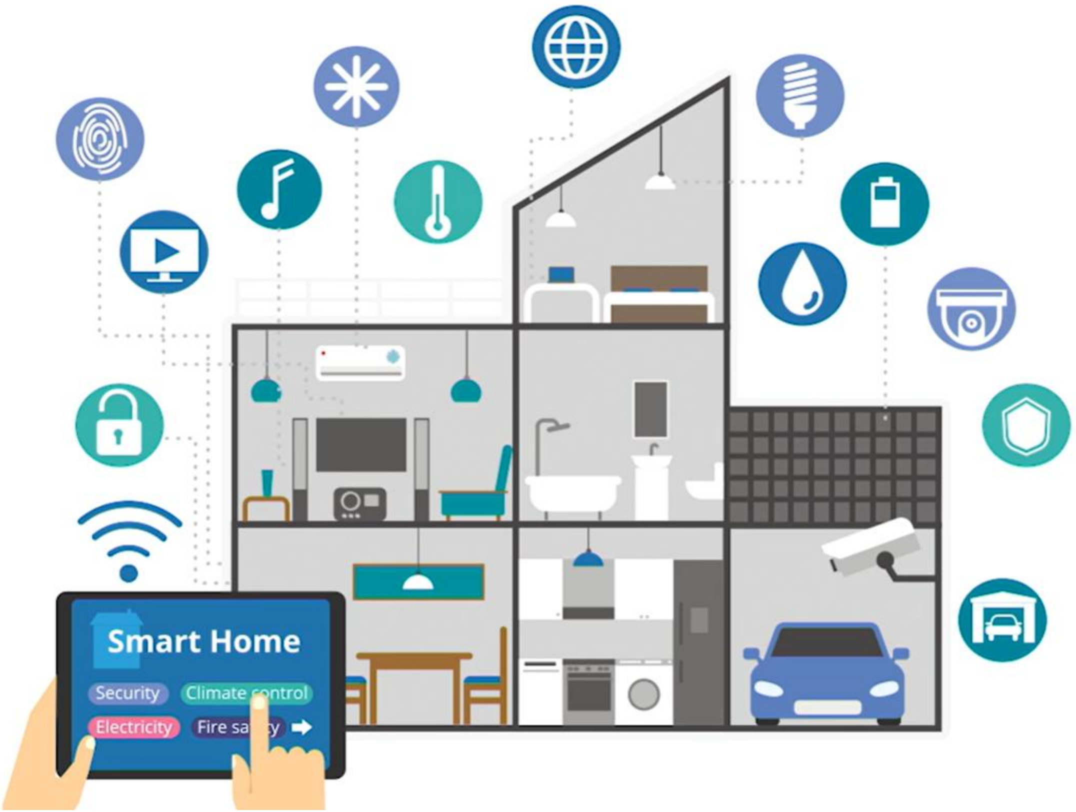
System Integration in a Layered approach



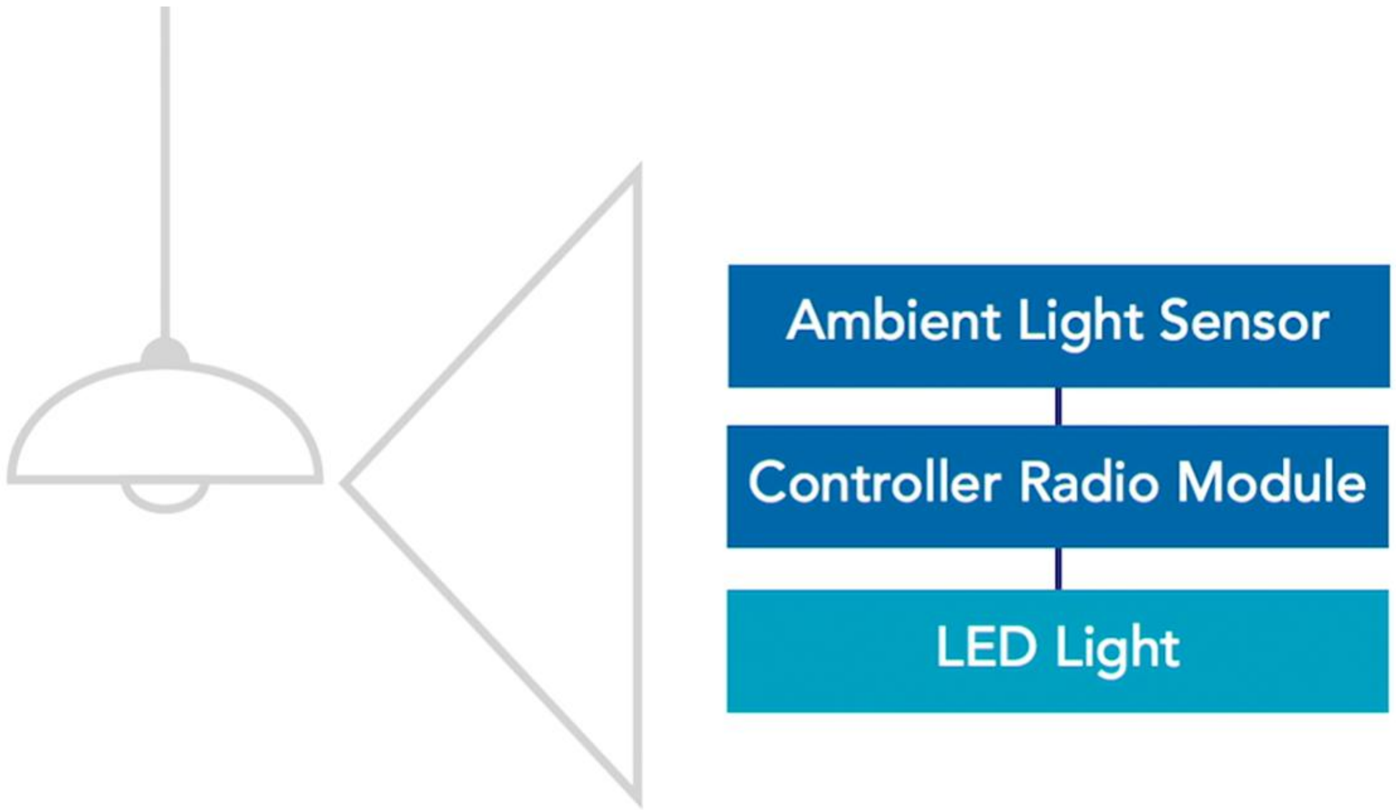
Light control system for smart home

- Design the smart light solution
- Develop the hardware for the end device
- Develop the software for the end device and application service

Case study



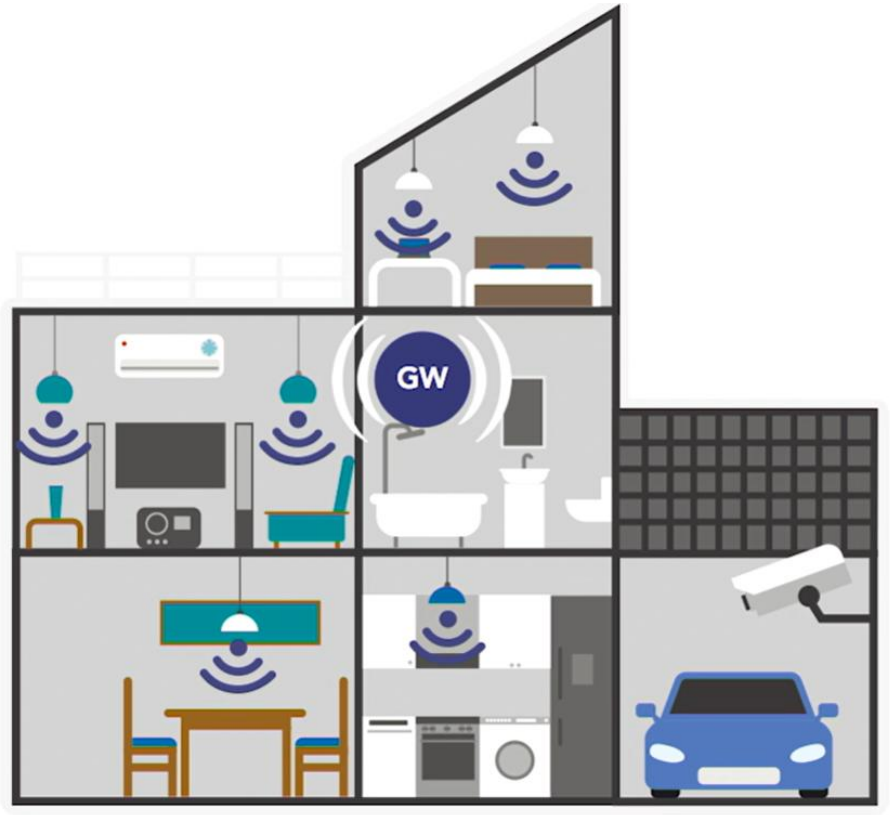
Case study



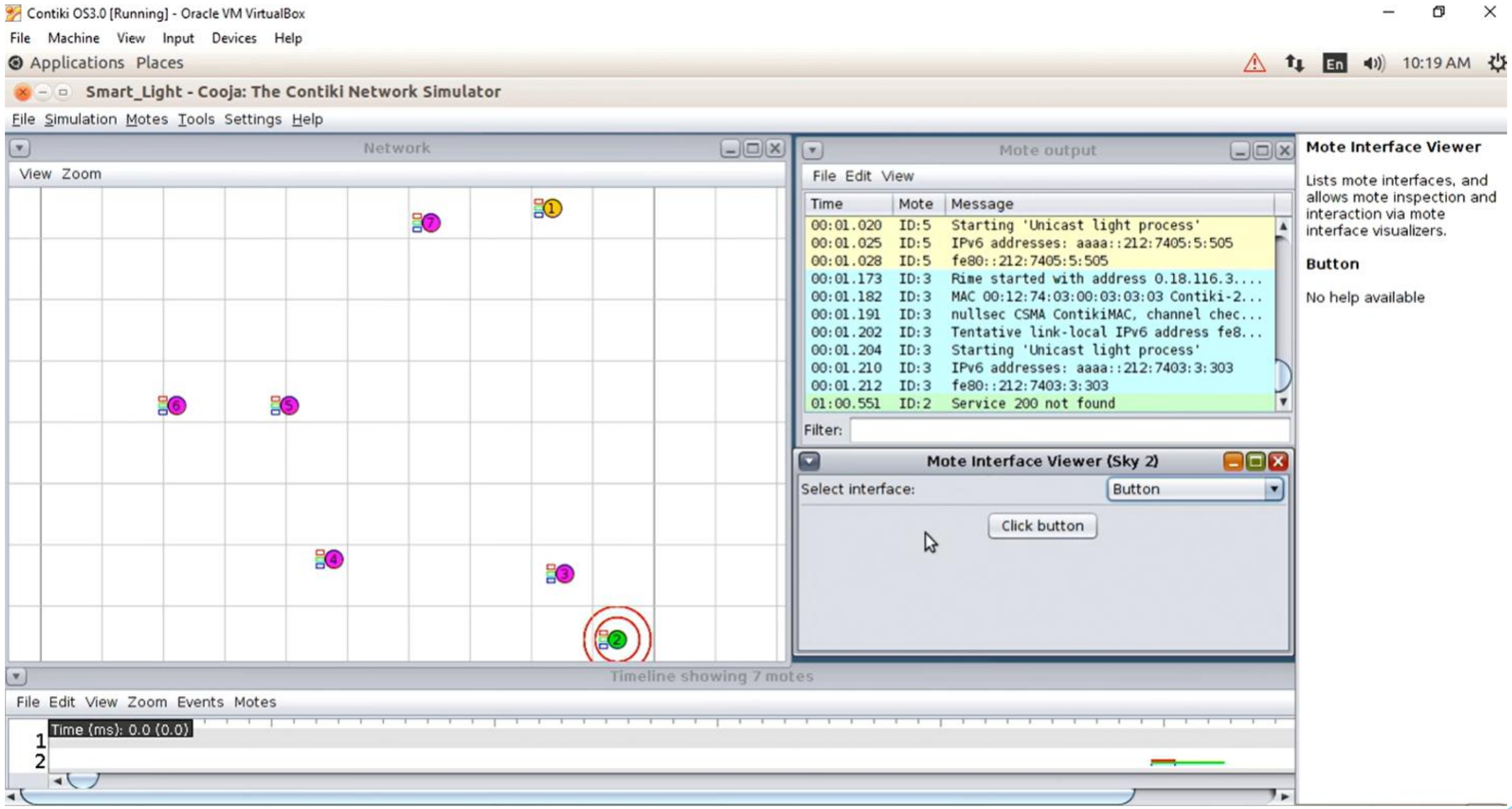
System requirement

Requirement	Value	Requirement	Value
Environment	Home, indoors	Battery life	Long
Network topology	Star, mesh	Latency	<100 ms
Mobility	Stationary	Radio frequency	Unlicensed 2.4 GHz
Traffic	Unidirectional low-rate infrequency data	Security	Normal
Data usage	Low (<50 bytes/s)		

Case study



Case study


Contiki OS3.0 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Applications Places
10:19 AM
Smart_Light - Cooja: The Contiki Network Simulator
File Simulation Motes Tools Settings Help
Network
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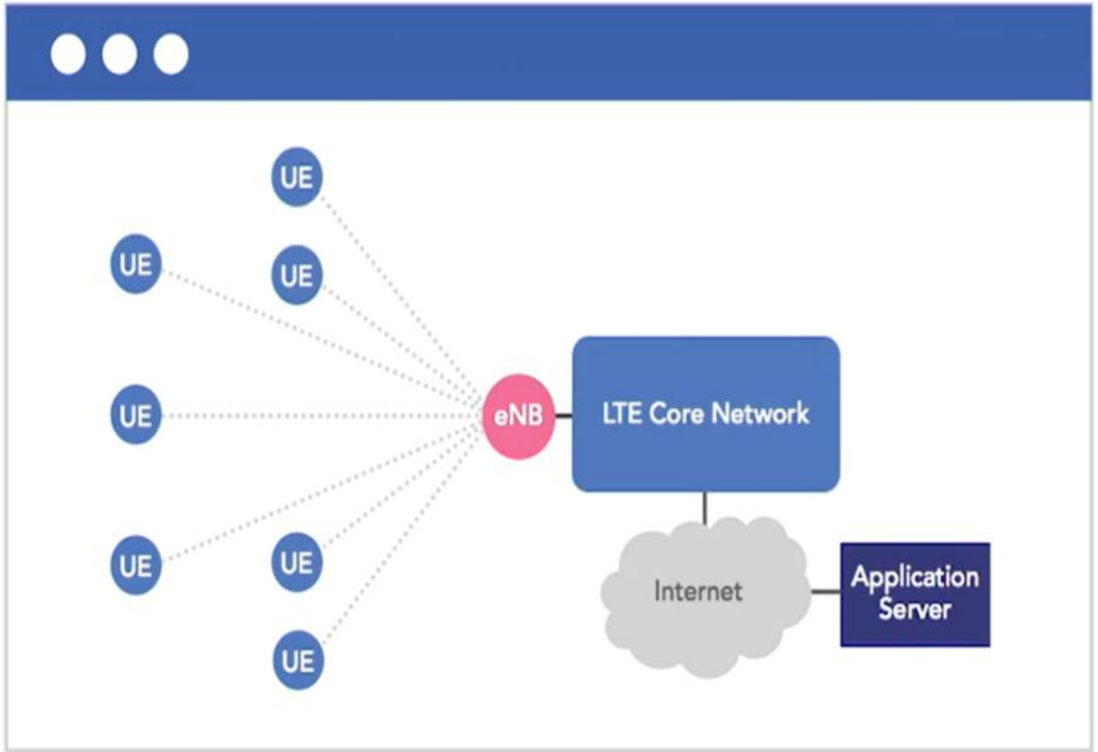
00:01.020 ID:5 Starting 'Unicast light process'
00:01.025 ID:5 IPv6 addresses: aaaa::212:7405:5:505
00:01.028 ID:5 fe80::212:7405:5:505
00:01.173 ID:3 Rime started with address 0.18.116.3....
00:01.182 ID:3 MAC 00:12:74:03:00:03:03:03 Contiki-2...
00:01.191 ID:3 nullsec CSMA ContikiMAC, channel chec...
00:01.202 ID:3 Tentative link-local IPv6 address fe8...
00:01.204 ID:3 Starting 'Unicast light process'
00:01.210 ID:3 IPv6 addresses: aaaa::212:7403:3:303
00:01.212 ID:3 fe80::212:7403:3:303
01:00.551 ID:2 Service 200 not found
Filter:
Mote Interface Viewer (Sky 2)
Select interface:
Button
Click button
Mote Interface Viewer
Lists mote interfaces, and allows mote inspection and interaction via mote interface visualizers.
Button
No help available
Timeline showing 7 motes
File Edit View Zoom Events Motes
Time (ms): 0.0 (0.0)

Case study 2

Case study 2


UE
UE
UE
UE
UE
UE
UE
eNB
LTE Core Network
Internet
Application Server


UE
Smart Street Light
LTE-M Modem
Light Controller Module
Street Light

Pushing data to the cloud

- Execute AT Commands provided by an LTE-M Module
- Establish an HTTP connection between LTE-M module and the cloud service
- Send status data to the cloud service

5G technology (fifth-generation)

5G is not just a faster version of 4G, but a transformative technology.

Its speed, low latency, and massive connectivity will drive future innovations.

Key features of 5G

Key features of 5G

- Greater Capacity:

5G can handle a much larger number of connected devices simultaneously. This is essential for the Internet of Things (IoT), where billions of devices are expected to be connected.

- Network Slicing:

which enables the creation of virtual networks tailored to specific applications. This allows for optimized performance for different use cases.

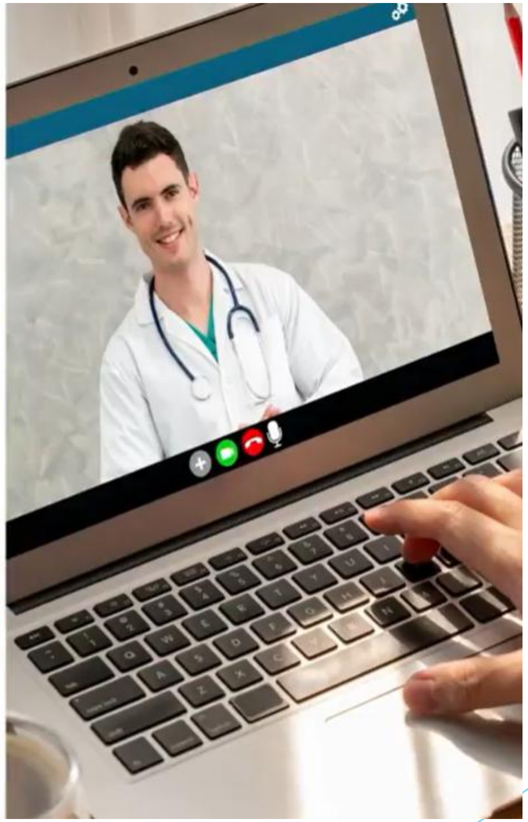
- Millimeter Wave (mmWave):

5G utilizes mmWave frequencies, which offer high bandwidth but have a shorter range and are more susceptible to obstacles.

- Beamforming:

5G uses beamforming technology to focus radio signals directly towards devices, improving signal strength and efficiency.

Application area



Application area

Enhanced Mobile Broadband:

Faster downloads and uploads, improved streaming, and enhanced mobile gaming.

Internet of Things (IoT):

Enables the connection of billions of devices, supporting smart cities, industrial automation, and remote monitoring.

Autonomous Vehicles:

Low latency and high reliability are crucial for autonomous vehicles to communicate with each other and with infrastructure.

Industrial Automation:

5G supports real-time control of industrial processes, improving efficiency and productivity.

Challenges implementing 5G

Challenges implementing 5G

- Infrastructure Deployment:

Deploying 5G infrastructure requires significant investment in new towers, small cells, and fiber optic cables.

- Coverage Limitations:

mmWave frequencies have limited range and are easily blocked by obstacles, requiring a dense network of small cells.

- Security Concerns:

The increased number of connected devices and the complexity of 5G networks raise security concerns.

- Power consumption:

Early 5G devices and towers used a lot of power. Technology is improving however.

Key ways AI is transforming 5G and IoT

AI in telecom

- Analyzes data from network components to predict failures.
- Allows to perform maintenance proactively.
- Reduces downtime and improves network reliability.

Impact of AI on 5G and IoT

Autonomous Vehicles:

- Makes self-driving cars a reality by combining 5G's low latency, IoT sensors, and AI decision-making.

Healthcare:

- Enables personalized medicine by analyzing data from wearable IoT devices.

Manufacturing:

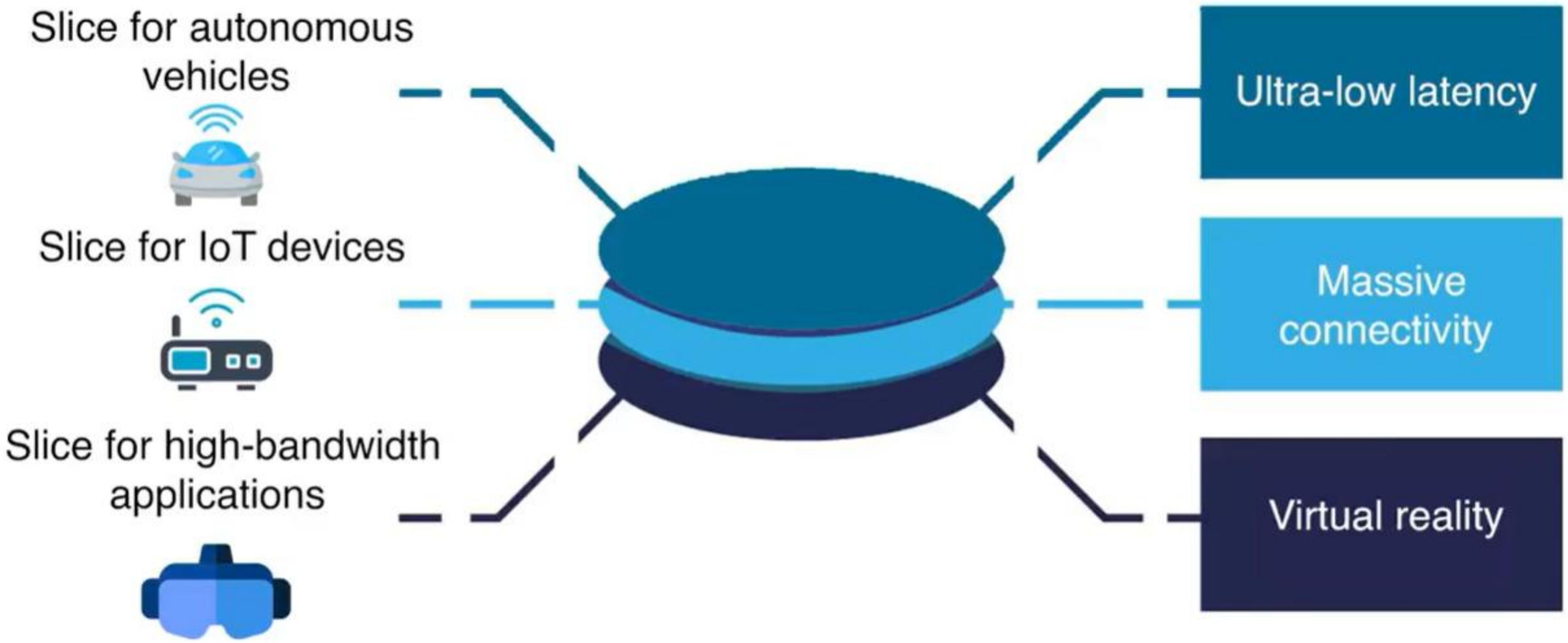
- Optimizes production lines in real time, increases efficiency, and reduces waste.

Challenges of AI in 5G and IoT

- **Data Privacy:** Concerns personal information gathered by IoT devices.
- **Algorithmic Bias:** Reflects societal prejudices from historical data.
- **Black Box Nature:** Complicates decision explanations.

AI Driven Innovations in 5G and IoT

5G Network Slicing

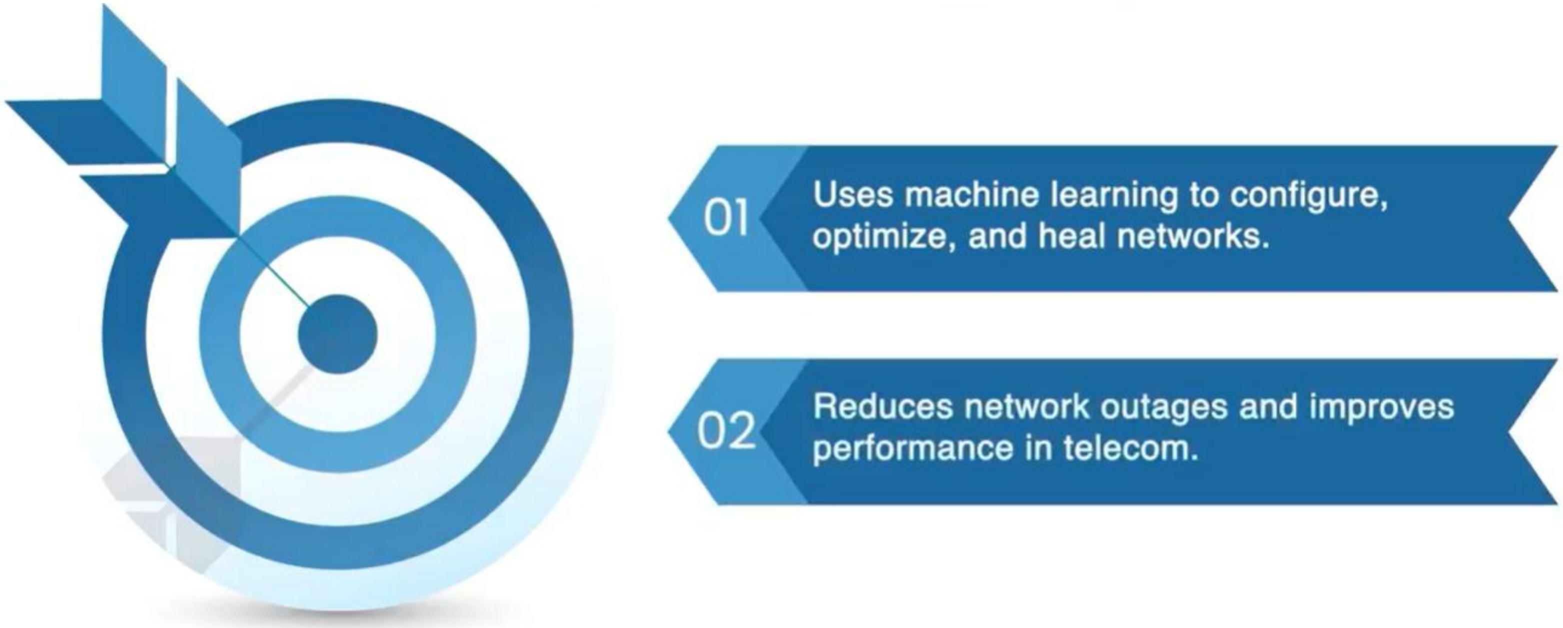


AI Driven Innovations in 5G and IoT

AI Driven Innovations in 5G and IoT

AI Driven Innovations in 5G and IoT
Edge AI Innovations challenges

AI Driven Innovations in 5G and IoT
Self-Organizing Network (SON)



AI Driven Innovations in 5G and IoT

Self-Organizing Networks: Challenges

- Make decisions that seem counterintuitive to human operators.
- Enhance the explainability of AI-driven network decisions to build confidence.

AI Driven Innovations in 5G and IoT

01 Transforms value extraction from IoT data.

02 Analyzes data from IoT sensors to optimize traffic flow and energy usage.

AI Driven Innovations in 5G and IoT

- Data generation by IoT devices can overwhelm analytics systems.
- Need to develop better data filtering and prioritization techniques.

AI Driven Innovations in 5G and IoT

References

- 1. IoT – Low power wireless networking

Question

Thank you